

Checkout the canteen

If...



- If the canteen serves 2000 students every day,
- how can the billing be automated without requiring data entry by staff?

Describe

- You are asked to design an automated system for calculating the cost of student-service meals at the University canteen.
- The system uses a camera to capture images of the trays containing the students' chosen food, then identifies the dishes and calculates the corresponding price.
- Identify, automatically crop the image of each detected dish and save it to a separate folder for the next recognition step.
- Identify the dishes using a Convolutional Neural Network (CNN) model to recognize the names of each dish that was cut in the previous step.
- The system needs to compare the names with a list of more than 15 popular dishes sold at the lunch canteen (e.g., braised pork, fried eggs, stir-fried water spinach, tofu, fried chicken, braised fish, cabbage soup, etc.).
- After identifying each dish, print the results to the console screen, along with the price of each dish and the total bill.



Price

STT	Món ăn	Giá tiền	Ghi chú
0	Cơm trắng	10.000	Một giá tiền nhiều hay ít
1	Đậu hũ sốt cà	25.000	
2	Cá hủ kho	30.000	
3	Thịt kho trứng	30.000	Một trứng, thêm 1 trứng + 6.000 đồng
4	Thịt kho	25.000	Không có trứng
5	Canh chua có cá	25.000	
6	Canh chua không cá	10.000	
7	Sườn nướng	30.000	
8	Canh rau	7.000	Cải hay muống
9	Rau xào	10.000	Lagim/củ sắn/đậu que/đậu đũa
10	Trứng chiên	25.000	Trứng chiên thịt

Crop image

Ảnh gốc



Ảnh đã cắt



Ảnh gốc

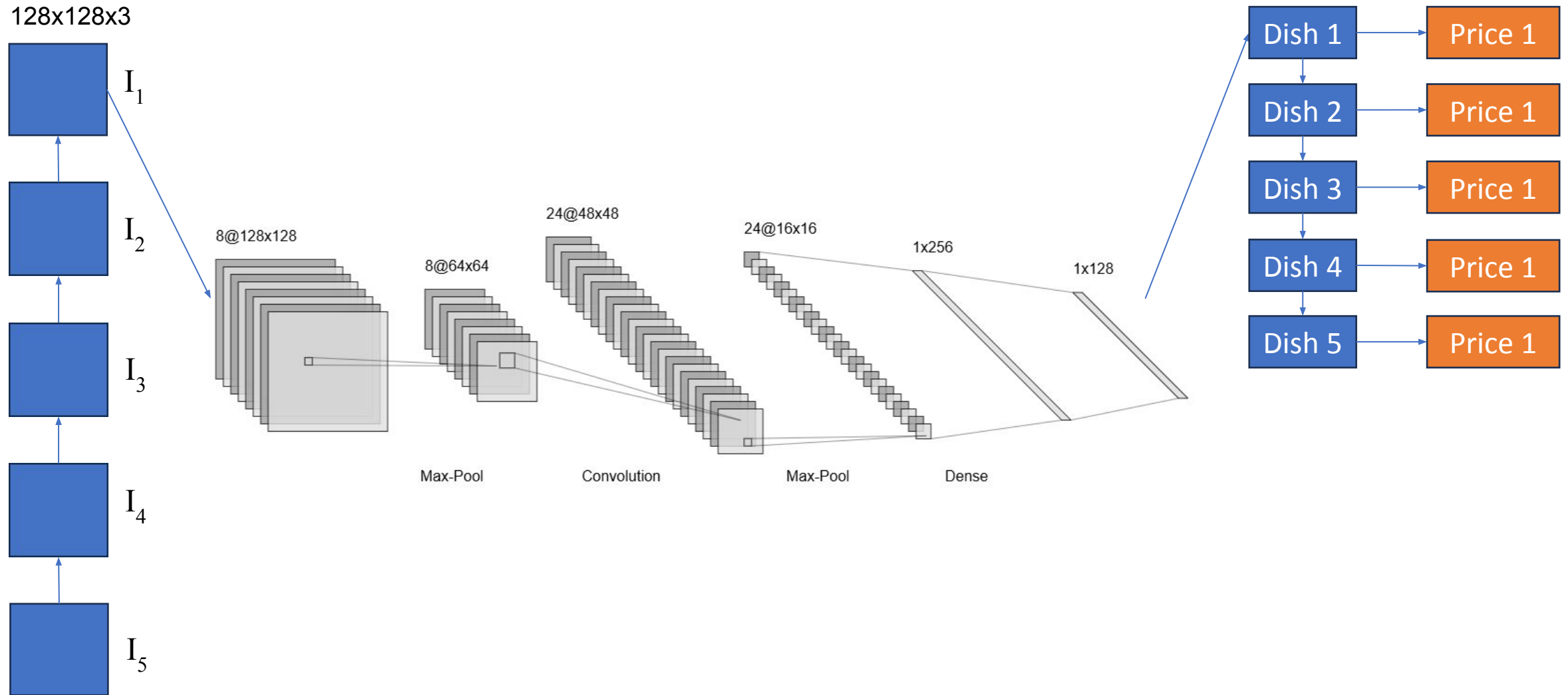


Ảnh đã cắt



```
from matplotlib import pyplot as plt
import matplotlib.image as mpimg
img = mpimg.imread('/content/IMG_1823.jpg')
x, y, w, h = 125, 135, 1428-125, 1325-135
# Crop (numpy slicing: img[y:y+h, x:x+w])
cropped = img[y:y+h, x:x+w]
# Show origin and cropped images
plt.figure(figsize=(10,5))
plt.subplot(1,2,1)
plt.title("Ảnh gốc")
plt.imshow(img)
plt.axis("off")
plt.subplot(1,2,2)
plt.title("Ảnh đã cắt")
plt.imshow(cropped)
plt.axis("off")
plt.show()
# Save a cropped image
plt.imsave("image1.jpg", cropped)
print("Đã lưu ảnh đã cắt thành công: image1.jpg")
```

CNN model



Dataset

Each class needs 200-500 images

dataset/

train/

ca_hu_kho/

thit_kho/

thit_kho_trung/

rau_xao/

canh_rau/

...

val/

test/